

Reproducible Manuscripts and Reports

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Reproducible Manuscripts and Reports

What is a Reproducible Report?

A **reproducible report** is a document that integrates data analysis, visualizations, and narrative text into a single cohesive output, which can be automatically regenerated and verified.

Key Components

- **Data:** Clearly sourced, documented, and accessible.
- **Code:** Transparent and executable by anyone.
- **Narrative:** Explanation and interpretation clearly linked to outputs.

Why Use Reproducible Reports?

- **Transparency:** Enhances trust and facilitates peer review.
- **Efficiency:** Easy updates when data or methods change.
- **Collaboration:** Simplifies teamwork and project continuity.

Tools for Creating Reproducible Reports

Open source:

- **Quarto:** Integrates R, Python, Julia and other languages directly into documents.
- **Jupyter Notebooks:** Supports Python and other languages for interactive analysis.

Output types

- Articles, presentations, dashboards, websites, blogs, and books in HTML, PDF, MS Word, **MS Powerpoint**, ePub, and more

Pdf **Beamer** **Powerpoint**

```
1 title: "My Document"
2 author: "Nicolas Reigl"
3 format:
4   pdf:
5     toc: true
6   number-sections: true
7   colorlinks: true
```

Authoring (with Quarto)

Syntax is similar in Jupyter notebooks

Project: `_quarto.yml`

`_quarto.yml`

```
1 project:
2   type: manuscript
3
4 manuscript:
5   article: index.qmd
6
7 format:
8   html:
9     comments:
10       hypothesis: true
11     docx: default
12
13   # (other formats)
14   # pdf: default
15
16 execute:
17   freeze: true
```

R and Python Chunks in Quarto

This is an example of an R chunk in Quarto. It fetches data from the Eurostat database using the `rdbnomics` package.

```
1 # Example RMarkdown chunk
2 library(rdbnomics) #
3 unemp <- rdb(ids = "Eurostat/ei_lmhr_m/M.PC_ACT.SA.LM-UN-T-TOT.EE") # fetch data
```